

Full Stack Development with MERN

Project Documentation format

1. Introduction :

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:**

Name	Role
Abinaya S	Team Leader
Irene Rachel	Team Member
Ayesha Mariyam S	Team Member
Jayasri J	Team Member

2. Project Overview :

- **Purpose:** The project aims to automate the **standard laptop procurement process** using the ServiceNow platform. The system allows employees to submit a Standard Laptop request through the Service Catalog.

After submission, the request is sent for approval. Once approved, **Flow Designer automatically creates a Catalog Task and assigns it to the Hardware assignment group**, reducing manual effort and improving request tracking.

- **Features:** Standard Laptop request through Service Catalog
 - Request validation and submission
 - Approval and rejection workflow
 - Automated Flow Designer trigger
 - Automatic Catalog Task creation
 - Automatic assignment to **Hardware** group
 - Laptop requirement details in the task
 - Request and task status tracking
 - Hardware task completion
 - Improved visibility of the procurement process

3. Architecture :

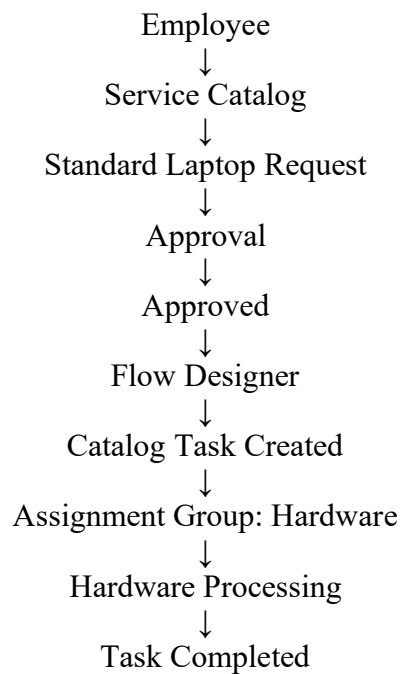
- **Frontend:** The project does not use a separate React frontend.

The user interface is provided by the **ServiceNow platform**, primarily through:

- Service Catalog
- Standard Laptop Catalog Item
- My Requests / Requested Items
- Approval interface
- Catalog Task interface

• **Backend:** The project does not use a separate Node.js or Express.js backend.

The application logic is implemented using **ServiceNow Flow Designer**.



• **Database:** The project uses the **ServiceNow platform's built-in database** to store:

- Requests
- Requested Items
- Approval records
- Catalog Tasks
- User information
- Task status
- Assignment information

No separate MongoDB database is used.

4. Setup Instructions :

- **Prerequisites:**

- Internet connection
- Web browser
- ServiceNow Developer Instance
- ServiceNow account
- Access to Service Catalog
- Flow Designer access
- Appropriate configuration/admin permissions

• **Installation:** Since this is a ServiceNow application, there is no traditional git clone, npm install, or MongoDB setup.

Steps

1. Log in to the ServiceNow Developer Instance.
2. Open **Service Catalog**.
3. Create/configure the **Standard Laptop** catalog item.
4. Configure the required laptop request fields.
5. Configure the approval process.
6. Open **Flow Designer**.
7. Create the automation flow.
8. Configure the flow trigger for an approved request.
9. Add an action to create a **Catalog Task**.
10. Configure the task details.
11. Set **Assignment Group = Hardware**.
12. Activate the flow.
13. Submit a Standard Laptop request to test the workflow.

5. Folder Structure :

Server:

Service Catalog

Used by employees to submit laptop requests.

Approval

Handles approval or rejection of the submitted request.

Flow Designer

Automates the creation of the Catalog Task after approval.

Catalog Task

Contains the laptop procurement/configuration task.

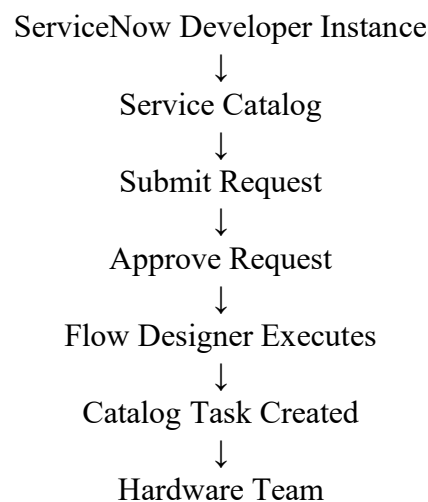
Hardware Assignment Group

Receives and processes the generated task.

6. Running the Application :

To Run

1. Log in to the ServiceNow Developer Instance.
2. Navigate to **Service Catalog**.
3. Select **Standard Laptop**.
4. Enter the required information.
5. Submit the request.
6. Open the submitted request.
7. Approve the request.
8. Flow Designer automatically executes.
9. Open the Requested Item.
10. Verify that a Catalog Task has been created.
11. Verify that the Assignment Group is **Hardware**.
12. Hardware team processes and completes the task.



7. API Documentation :

The project does **not expose custom REST APIs or backend endpoints**.

The automation is implemented using ServiceNow's native platform functionality.

Component	Purpose
Service Catalog	Submit laptop request
Requested Item	Store submitted request

Component	Purpose
Approval	Approve/reject request
Flow Designer	Automate workflow
Catalog Task	Create procurement/configuration task

Assignment Group Route task to Hardware

External API :

No external API is used.

8. Authentication :

Authentication is handled by the **ServiceNow platform**.

Users log in using their authorized ServiceNow credentials.

Authorization and access are controlled using ServiceNow's built-in:

- User roles
- Permissions
- Access Controls
- Assignment Groups

The project does not implement custom JWT authentication, sessions, or OAuth-based authentication.

9. User Interface :

Main UI Screens

1. Service Catalog
2. Standard Laptop Request Form
3. Submitted Request
4. Approval Screen
5. Requested Item
6. Catalog Task
7. Hardware Assignment
8. Task Completion / Status

TC-001 – Laptop Request Submission

Employee selects the Standard Laptop and submits the required information.

TC-002 – Request Approval

The approver reviews the submitted request and approves it.

TC-003 – Automatic Catalog Task

After approval, Flow Designer automatically creates the Catalog Task.

TC-004 – Task Details

The generated task contains the required laptop details and task description.

TC-005 – Hardware Assignment

The generated task is assigned to the **Hardware** group.

10. Testing :

Testing was performed on the ServiceNow workflow to verify that the complete procurement process functions correctly.

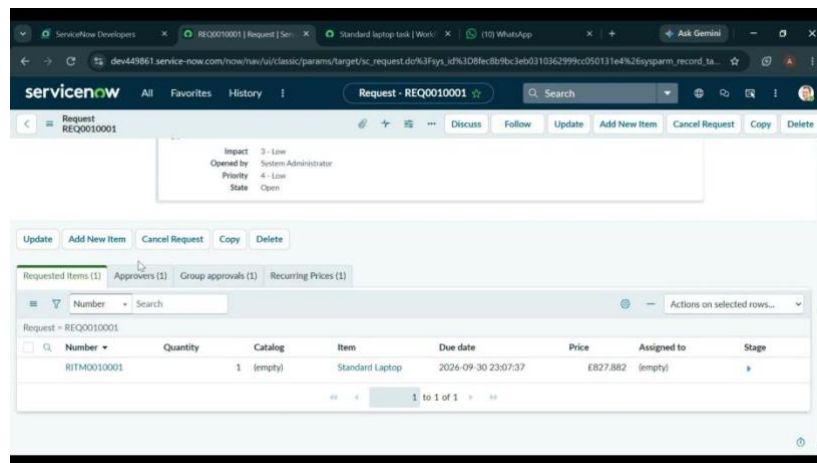
Testing Types

- Functional Testing
- Workflow Testing
- Approval Testing
- Automation Testing
- Assignment Testing
- Status Tracking Testing
- Performance Testing
- User Acceptance Testing

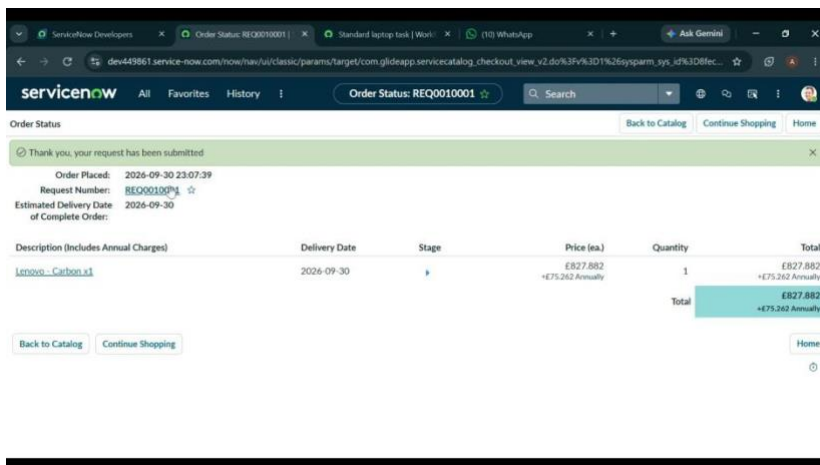
Functional Test Cases

Test Case	Description	Expected Result	Status
TC-001	Submit Standard Laptop request	Request created	Pass
TC-002	Approve laptop request	Request approved	Pass
TC-003	Automatic Catalog Task creation	Task automatically created	Pass
TC-004	Verify task description	Required details populated	Pass
TC-005	Verify Hardware assignment	Task assigned to Hardware	Pass
TC-006	Reject laptop request	Request rejected	Pass
TC-007	Track request status	Status displayed correctly	Pass
TC-008	Complete Hardware task	Task completed	Pass

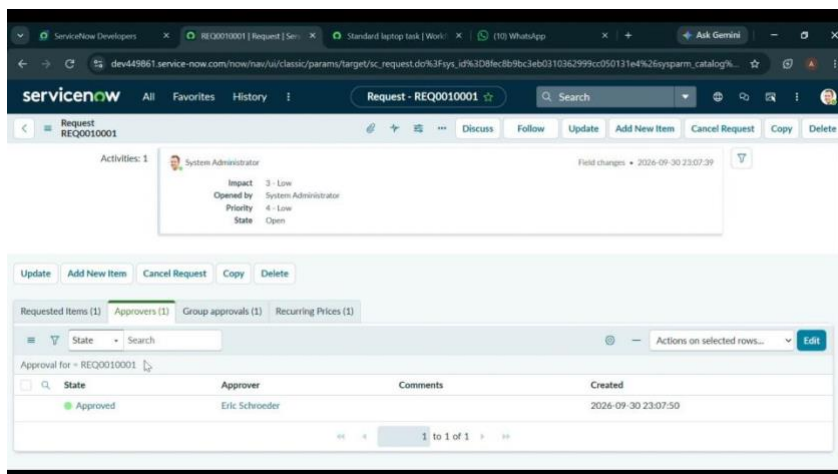
11. Screenshots or Demo :



This screenshot shows the 'Request - REQ0010001' page in ServiceNow. The page header includes the ServiceNow logo and navigation tabs. A search bar is present. Below the header, there are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. A dropdown menu is open, showing 'Impact: 3 - Low', 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. Below this, there are tabs for 'Requested Items (1)', 'Approvers (1)', 'Group approvals (1)', and 'Recurring Prices (1)'. The 'Requested Items (1)' tab is selected, showing a table with one item: 'RITM0010001' with a quantity of 1, catalog 'Standard Laptop', due date '2026-09-30 23:07:37', price '£827.882', and assigned to '(empty)'. The table has columns for 'Number', 'Quantity', 'Catalog', 'Item', 'Due date', 'Price', 'Assigned to', and 'Stage'.



This screenshot shows the 'Order Status - REQ0010001' page in ServiceNow. The page header includes the ServiceNow logo and navigation tabs. A search bar is present. Below the header, there are buttons for 'Back to Catalog', 'Continue Shopping', and 'Home'. A green message bar states 'Thank you, your request has been submitted'. Below this, the order details are displayed: 'Order Placed: 2026-09-30 23:07:39', 'Request Number: REQ0010001', and 'Estimated Delivery Date of Complete Order: 2026-09-30'. A table shows the order items with columns: 'Description (Includes Annual Charges)', 'Delivery Date', 'Stage', 'Price (ex.)', 'Quantity', and 'Total'. The table has one item: 'Lenovo - Carbon x1' with a delivery date of '2026-09-30', stage 'Standard Laptop', price '£827.882', quantity '1', and total '£827.882'. A 'Total' row shows the total price '£827.882'. Below the table, there are buttons for 'Back to Catalog', 'Continue Shopping', and 'Home'.



This screenshot shows the 'Request - REQ0010001' page in ServiceNow, specifically the 'Approvers (1)' tab. The page header includes the ServiceNow logo and navigation tabs. A search bar is present. Below the header, there are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. A dropdown menu is open, showing 'Impact: 3 - Low', 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. Below this, there are tabs for 'Requested Items (1)', 'Approvers (1)', 'Group approvals (1)', and 'Recurring Prices (1)'. The 'Approvers (1)' tab is selected, showing a table with one approver: 'Approved' with an approver of 'Eric Schroeder' and a created date of '2026-09-30 23:07:50'. The table has columns for 'State', 'Approver', 'Comments', and 'Created'.

ServiceNow Request - REQ0010001

Requested Item RITM0010001

Number: RITM0010001

Item: Standard Laptop

Request: REQ0010001

Requested for: System Administrator

Due date: 2026-09-30 23:07:37

Configuration Item:

Watch list:

Opened: 2026-09-30 23:07:38

Opened by: System Administrator

Stage: Request Approved

State: Open

Quantity: 1

Estimated delivery:

Backordered:

Order Guide:

Variables:

Optional Software:

- ☐ Adobe Acrobat
- ☐ Adobe Photoshop

Additional software requirements:

ServiceNow Catalog Task - SCTASK0010001

Number: SCTASK0010001

Assigned to:

Configuration Item:

Active: ☒

Short description: Laptop need to configured

Description: Laptop need to configured

Work notes: Work notes

Approval: Approved

Priority: None

State: Open

Request Item: RITM0010001

Requested for: System Administrator

Comments (Customer visible):

Post

Activities: 1

Field changes: 2026-09-30 23:11:28

12. Known Issues :

- The workflow depends on correct ServiceNow configuration.
- Users require appropriate permissions to access the Service Catalog and workflow records.
- Flow execution may take a short amount of time after approval.
- Incorrectly configured approval conditions may prevent the automation from triggering.
- Incorrect assignment-group configuration can prevent the task from reaching the Hardware team.
- The project currently focuses on **Standard Laptop procurement** and does not cover all IT equipment categories.

13. Future Enhancements :

The project can be extended with:

1. **Multiple IT Equipment Categories**
Add desktop, monitor, keyboard, mouse, printer and other equipment requests.
2. **Automated Notifications**
Send email or ServiceNow notifications when a request is submitted, approved, rejected or completed.
3. **Request Dashboard**
Provide an administrative dashboard showing pending, approved, rejected and completed requests.
4. **SLA Management**
Add SLA tracking for procurement and hardware task completion.
5. **Automated Escalation**
Escalate requests that remain pending beyond the defined time.
6. **Inventory Integration**
Integrate laptop inventory information to check available devices.
7. **Approval Hierarchy**
Add multiple approval levels for higher-value equipment.
8. **Additional Catalog Items**
Extend the same Flow Designer approach to other IT procurement processes.